Photo conductivity :-

The conductivity of semiconductor due to photon absorption is called photo conductivity. When a photon having energy (suitable frequency) equal to energy gap in Semiconductor is absorbed, this optical absorption results to extra free conduction electron in conduction region and hence conductivity of semiconductor can be observed. By measuring the amount of photon gets absorbed of known wavelength, it can be used to measure the energy gap of the semiconductor.

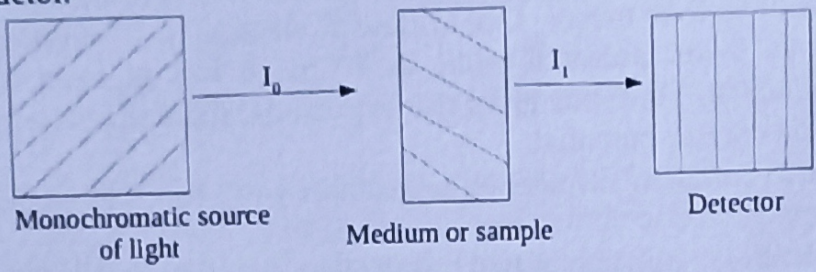
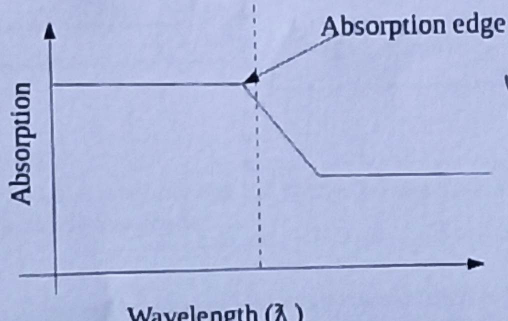


Fig: Experimental arrangement for the absorption of radiation by semiconductor sample.

A single wavelength light from monchoromatic light source of known intensity I_0 is allowed to incident in Semiconductor and finally I_1 intensity of transmitted intensity was measured with detector.

Hence, the absorbed intensity I_0-I_1 of particular wavelength is obtained. The variation of absorption with wavelength of light shows , on increasing the wavelength of light initially the absorbance is larger and further after particular wavelength known as absorption edge , it starts to fall showing that energy photon is less than band gap and reduction in absorption. On further increasing wavelength for longer wavelength , there is poor absorption as shown in fig.



by S.G sir

Fig: Variation of absorption of light by Semiconductor with its wavelength.

If the energy of incident photon is less than energy gap of Semi conductor, it is unable to excite electron from valence band to conduction band and hence no more radiation is absorbed and pass through sample or to say medium is transparent to photon. The minimum frequency required for photo conductivity is

$$E_g = h\nu_c = \frac{hc}{\lambda_c} = \frac{6.63 X 10^{-34} X\ 3 X 10^{8}}{1.1 X 10^{-6}} = 1.8 X 10^{-19} J$$

where λ_c= 1.1 X10^{-6} m is edge wavelength for Si crystal.

Metal-Metal junction (The contact potential):

In metal the outermost occupied energy state at absolute temperature 0K is called Fermi energy level. The energy difference between Fermi energy level to conduction band is refer as work function.

For two metal conductors with work function ϕ_1 and ϕ_2 are connected together, such that $\phi_2 > \phi_1$. As electron at 1st metal is at higher energy about Fermi energy level than that of 2nd metal , electron at contact flows from 1st metal to 2nd. The flow of electron will continue until the highest occupied energy level is same in both metals. Due to flow of electron in 2nd metal it becomes negatively charged and hence have lower potential while as 1st metal loss electron it becomes positively charged and at higher potential. Thus, this gives rise to potential difference at a junction of two metal and this potential is called contact potential.

Fermi energy gets lower to common Fermi level with lower work function while it values increases in case of higher work function.

For V_c be the contact potential at junction of two metals, then

$$eV_c = \phi_2 - \phi_1$$

$$\text{or,}\quad V_c = (\phi_2 - \phi_1)/e$$

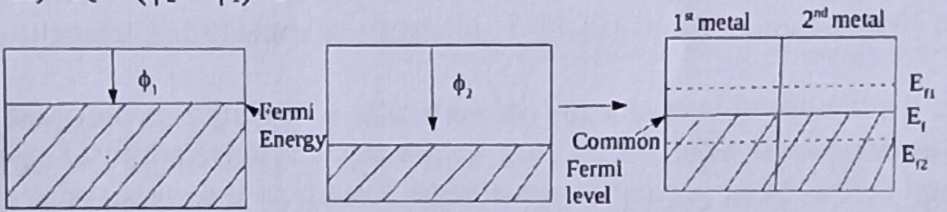


Fig: Metals contact potential

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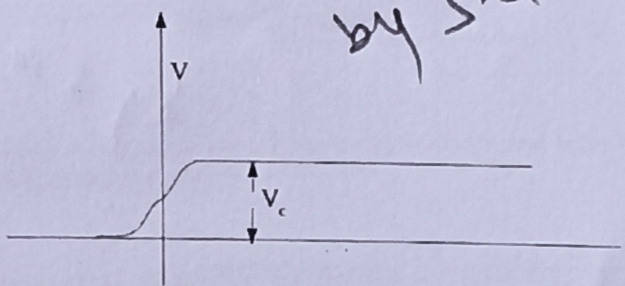

Fig: Shift in potential at metal contact by common contact potential.

The Semiconductor Diode:

A P-type extrinsic Semiconductor is one with deficiency of electron at +ve potential while a N-type extrinsic Semiconductor is rich in electron and at -ve potential. When two P-type and N-type Semiconductor are joined together, the electron from N-type jumps to P-type to certain extend about 10^{-6}m and a potential is developed across the junction called barrier potential. Thus, developed layer formed about junction is called depletion layer.

In order to flow of charge through the junction, a charge with greater potential than junction potential is required and hence this potential acts as a barrier so named barrier potential. The presence of depletion layer gives rise to junction and diffusion capacitance.

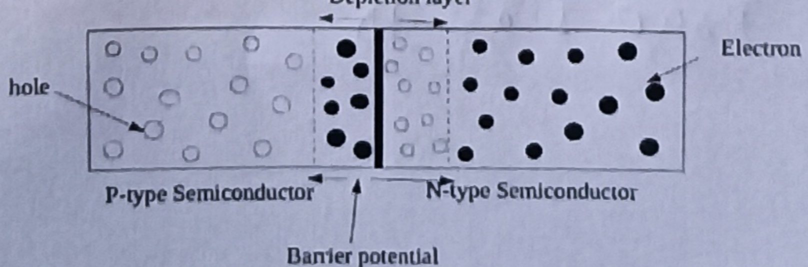

Fig:- a junction diode semiconductor.

P-N junction diode and Band scheme of a P-N junction diode:

A P-N junction diode is a electronic component formed with two P-type and N-type extrinsic semi-conductor in contact forming a junction. The P-region is called anode and N-region is called cathode hence named diode.

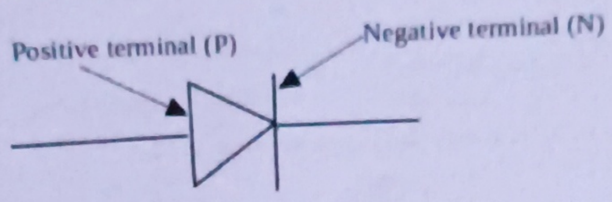

Fig: A symbolic representation of P-N junction diode

As P-region is electron deficient region compare to N-region. The contact of two regions results to diffusion of electron from N-side to P-side leaving hole behind in N-side. This implies a P-region -vely charged while N-side is +vely charged. Thus, a potential difference 'V_c' across the junction will be created and called it as junction potential or barrier potential.

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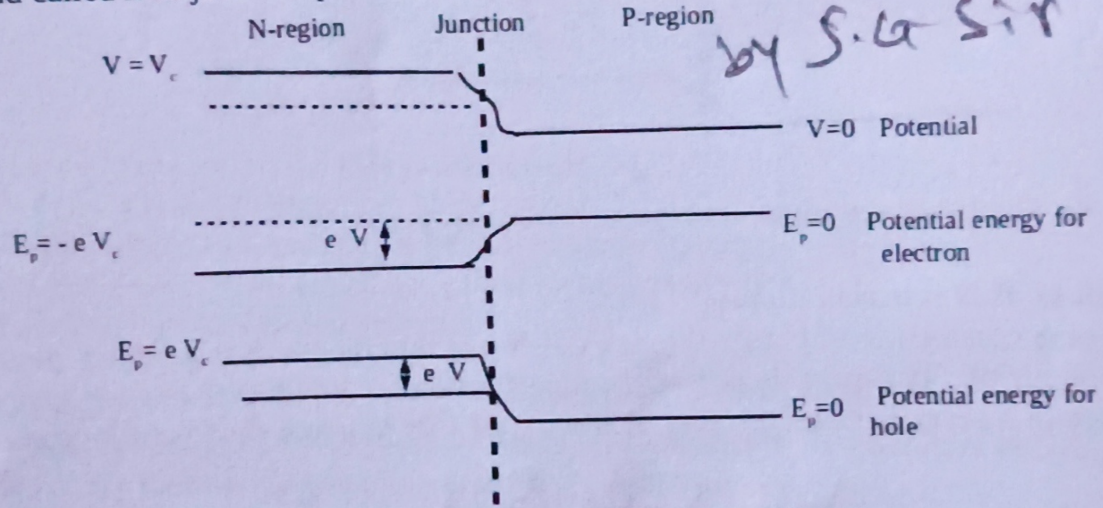


Fig: Variation of potential and P.E barrier for electrons and holes in in P-N junction diode.

A N-side is at higher potential compare to P-side as it is +vely charged further, a potential energy at a point is define as

$$W= q V$$

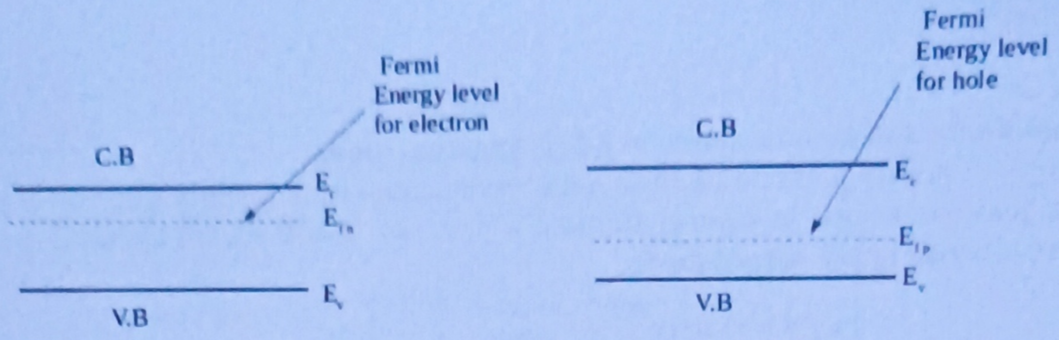
Therefore, for electron its potential energy at N-side is; $E_{p.E}=-|e|V_c$ while at P-side is equal to $E_{p.E}=0$ as V=0 here. Thus, for electron, it is with lower P.E. in N-region compare to that at P-region. This energy difference acts as barrier for flow of charge across diode.

Similarly, for hole its potential energy at P-side is equal to $E_{p.E}=0$ as V=0 here, while at N-side is $E_{p.E}=|e|V_c$. Thus, for hole, it is with higher P.E. in N-region compare to that at P-region. This energy difference acts as barrier for flow of hole across diode.

Hence, these P.E. barriers stops further flow of majority charge carriers across the junction. The shift of potential energy barrier at junction is equal to the same difference in Fermi energy level on both side of contact potential i.e.

potential energy barrier = difference between energy of conduction and valence band.

or, $e V_c = E_{fn} - E_{fp}$



For N-type semiconductor. For P-type semiconductor.

Fig: Fermi energy levels in N-type and P-type before in contact.

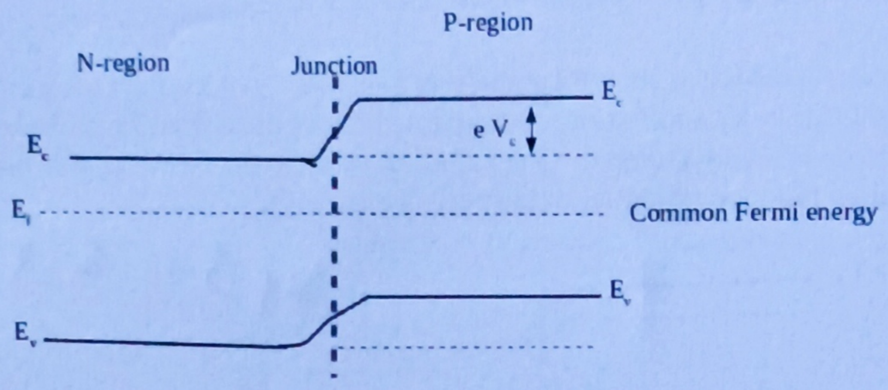


Fig: Energy band after contact of N-type and P-type semiconductor.

Biasing Circuit of P-N junction diode:-

For, a conduction of charge through a P-N junction diodes, it is required to overcome junction barrier potential. So, a suitable potential connection across P-N junction diode with help of an external source of battery is called biasing. A biasing of P-N junction diode can be done in two distinguish ways.

by S. G. Sir

1. Forward biasing:-

When N-type of a P-N junction diode is connected to -ve terminal and its P-type to +ve terminal of battery then such a connection is called forward biasing.

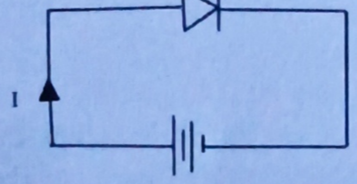


Fig: Forward biased of P-N junction diode.

When -ve terminal of battery is connected to N-side and +ve terminal to p-side of junction diode, it lower down potential for both electrons and holes at N-side and P-side of junction respectively by an amount equal to applied potential 'V' in comparison to their barrier potential 'V_c' initially. So, a new reduced potential across junction is V-V_c.

So, for electron its reduced potential energy at N-side is; $E_{p.E}=-|e|(V-V_c)$ while at P-side is equal to $E_{p.E}=0$.

Similarly, for hole its reduced potential energy at P-side is equal to $E_{p.E}=0$ while at N-side is $E_{p.E}=|e|(V-V_c)$.

Hence, as the potential barrier decreases on both sides of junction this results to decreases in potential energy barrier.. This lowering of energy barrier will greatly increase flow of majority charge carriers crossing across the junction giving rise to a large current through diode.

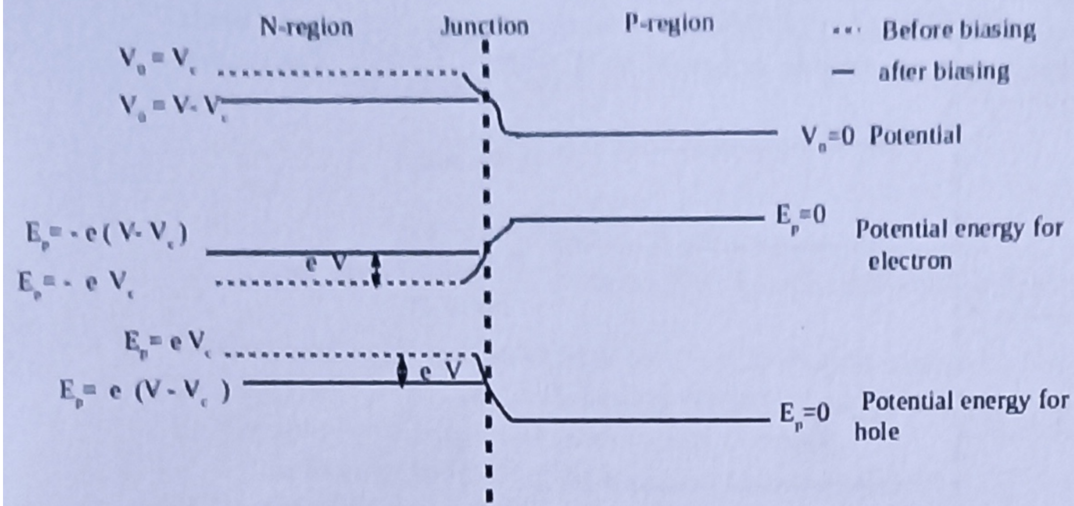


Fig: Variation of potential and P.E barrier for electrons and holes in forward biased condition of P-N junction diode.

by S.G Sir

The characteristic of forward biased P-N junction diode is,

1. flow of current is due to majority charge carriers.
2. the width of depletion layer and barrier potential decreases.
3. The diode offers very low resistance called forward resistance.
4. The diode behaves as a closed switch.

1. Reverse biasing:-

When N-type of a P-N junction diode is connected to +ve terminal and its P-type to -ve terminal of battery then such a connection is called reverse biasing.

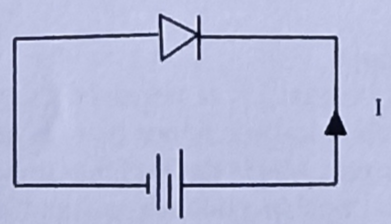


Fig: Reverse biased of P-N junction diode.

When +ve terminal of battery is connected to N-side and -ve terminal to p-side of junction diode, it will raise potential for both majority charge carrriers electrons and holes at N-side and P-side of junction respectively by an amount equal to applied potential 'V' in comparison to their barrier potential 'V_c' initially. So, a new raised potential across junction is $V+V_c$.
So, for electron its new increased potential energy at N-side is; $E_{p.E}=-|e|(V+V_c)$ while at P-side is equal to $E_{p.E}=0$.
Similarly, for hole its reduced potential energy at P-side is equal to $E_{p.E}=0$ while at N-side is $E_{p.E}=|e|(V+V_c)$.

Hence, as the potential barrier increases on both sides of junction this results to raising in potential energy barrier. This increase of energy barrier will greatly decrease flow of majority charge carriers crossing across the junction giving rise absent of current through diode.

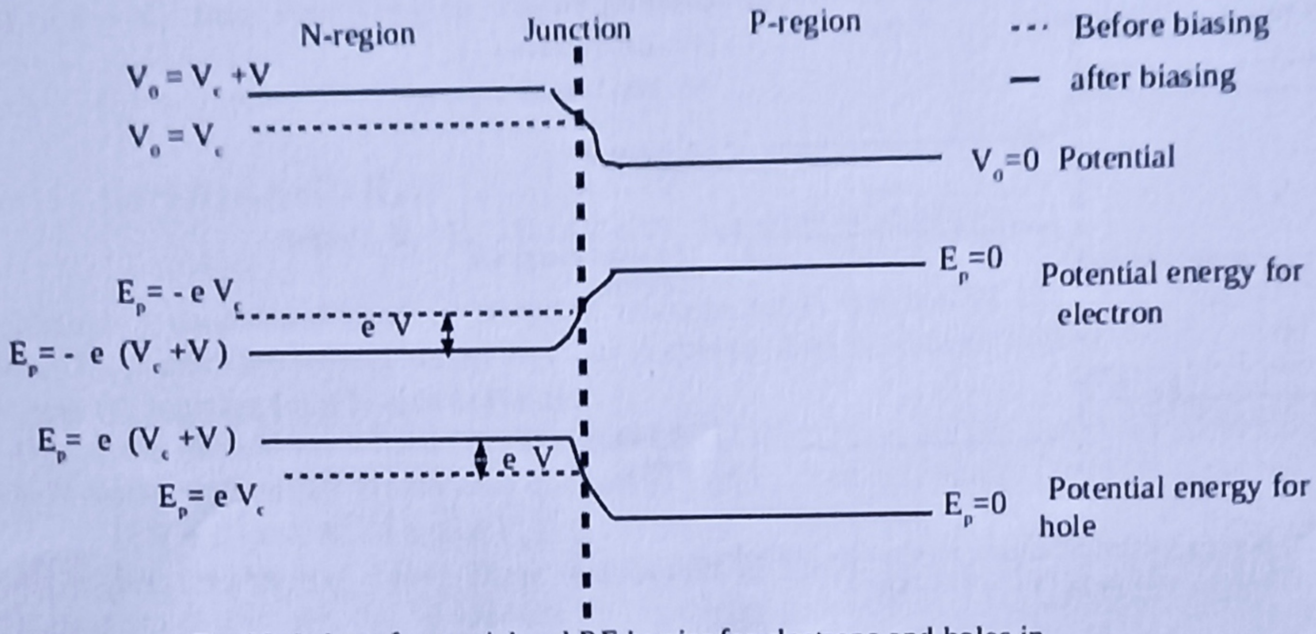


Fig: Variation of potential and P.E barrier for electrons and holes in reversed biased condition of P-N junction diode.

The characteristic of reverse biased P-N junction diode is,

1. The negligible current inside diode will flow due to minority charge carriers.
2. The width of depletion layer and barrier potential increases.
3. The diode offers very high resistance called backward resistance.
4. The diode behaves as an open switch.

by S.G. Sir

Current flow across the P-N junction diode:-

When P-type and N-type extrinsic semiconductor are kept at a contact, a P-N junction is formed with a potential barrier energy of $|e|V_c$, where V_c is barrier (contact) potential. The corresponding barrier energy checkout the flow of charge carriers across P-N junction.

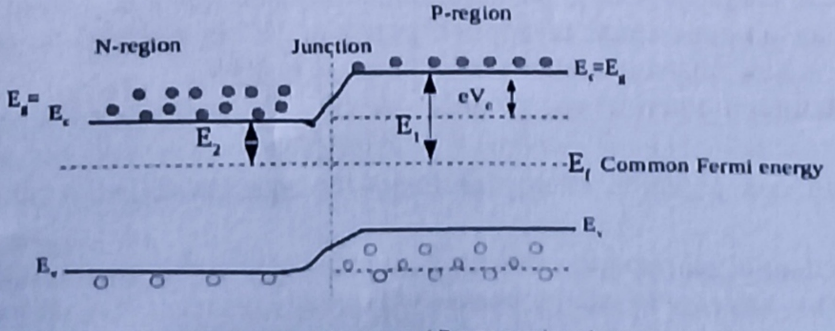
1. Equilibrium currents across the P-N junction:-

In an absence of external supply potential i.e. at unbaised case, equal number of electrons and holes continuously flowing in opposite direction and hence there is net zero flow of charge about junction. This stage is called equilibrium currents across the P-N junction.

Let, us consider the flow of electrons in conduction band of both N-type and P-type semiconductors. The number of electron (majority carriers) at conduction band in semiconductor is given by

$$N_e = N_c e^{-(E_g - E_f)} / K_B T \quad \text{.......................} \; 1$$

As (E_g - E_f) is much greater for P-type than for N-type as shown in fig. Below. So, the concentration of electron for N-type semiconductor is very larger than for P-type semiconductor.



$E_1 = E_2 + eV_C$

for P-region

$E_g - E_f = E_2 + eV_C$

for N-region

$E_g - E_f = E_2$

Fig: Energy band after contact of N-type and P-type semiconductor.

Now, as the electrons (minority carriers) at conduction band of P-region is at higher potential energy and hence are not oppose by barrier potential from crossing junction from P to N-region. So, the current $i(P \rightarrow N)$ from P to N-region will be proportional to total number of electrons in P-region.

$$i(P \rightarrow N) = A e^{-(E_g - E_f)} / K_B T$$

$$i(P \rightarrow N) = A e^{-E_1} / K_B T \quad \ldots\ldots\ldots\ldots\ldots\ldots 2$$

where, $E_1 = E_g - E_f$ of P-side, A is arbitrary constant.

Further, although a conduction band of N-region contains large number of electrons (majority carriers), only those electrons having energy equal to or greater than the barrier energy $|e|V_c$ will be able to cross the junction from N-side to P-side.

Hence, the current flows from N to P-region i.e. $i(N \rightarrow P)$ is proportional to number of electrons in N-region with energy greater than or equal to $|e|V_c$ and is given by,

$$i(N \rightarrow P) = A\, N_e\, f[E \geq |e|\, V_c] \quad \ldots\ldots\ldots\ldots\ldots\ldots 3$$

At higher temperature, Fermi Dirac distribution of electrons can be approximated by Maxwell Boltzmann distribution. So, for electron having energy greater than barrier energy $|e|V_c$, the distribution function is

$$f[E \geq |e|\, V_c] = e^{-|e| V_c / K_B T} \quad \ldots\ldots\ldots\ldots\ldots\ldots 4$$

Further, from equation 1 ; concentration of electron at conduction of N-region is,

$$N_e = 2\sqrt{\left(\frac{m_e K_B T}{2\pi\hbar^2}\right)^3}\, e^{-\frac{E_g - E_f}{K_B T}}$$

For small temperature 'T',

by J.G. Sir

$$N_e \approx e^{-\frac{E_g - E_f}{K_B T}}$$

$$N_e = e^{-\frac{E_2}{K_B T}} \quad \text{where, } E_2 = (E_g - E_f) \text{ for N-region.}$$

So, using eqn 5, 4 and 3; we get,

$$i(N \rightarrow P) = A\, e^{-\frac{E_g - E_f}{K_B T}}\, e^{-|e| V_c / K_B T}$$

$$i(N \rightarrow P) = A\, e^{-\frac{(E_2 + |e| V_c)}{K_B T}}$$

$$i(N \rightarrow P) = A\, e^{-\frac{E_1}{K_B T}} \quad \ldots\ldots\ldots\ldots\ldots\ldots 6$$

where, $E_1 = E_2 + |e|V_c$ is raised potential energy for hole in N-region.

From eqn 2 and 6; we get,

$$i(N \rightarrow P) = i(P \rightarrow N) = A\, e^{-\frac{E_1}{K_B T}}$$

Therefore, the flow of electron from N to P-region is equal to that from P to N-region. Thus, net current flowing across P-N junction is zero.

2. Diode equation (Net flow of charge carriers across the P-N junction diode):-

When an external biasing potential with power source is applied across diode in forward bias stage. The height of barrier potential energy at P-N junction is reduced to $|e|(V_c - V)$.

Then, the flow of electrons (majority carriers) from N to P region will be more than flow of electrons (minority carriers) from P to N-region. So, the net electron current from N to P-region or across junction in forward direction is,

$$i = i(N \to P) - i(P \to N) \quad \ldots\ldots\ldots\ldots\ldots\ldots 1$$

The flow of current from N to P region due to majority electron is proportional to the electrons in conduction band with energies greater than height of reduced barrier potential $|e|(V_c - V)$ and is given by,

$$i(N \to P) = A\,e^{-\frac{(E_2 + |e|(V_c - V))}{K_B T}} \quad \ldots\ldots\ldots\ldots\ldots\ldots 2$$

where, $E_2 = (E_g - E_f)$ for N-region.

Further, as the flow of electrons (minority carriers) from P to N region is same as in unbaised condition i.e. doesn't change with forward biasing potential.

$$i(P \to N) = A\,e^{-\frac{E_1}{K_B T}} \quad \ldots\ldots\ldots\ldots\ldots\ldots 3$$

where, $E_1 = (E_g - E_f)$ for P-region.

Then,from eqn 1, 2 and 3;

$$i = A\,e^{-\frac{(E_2 + |e|(V_c - V))}{K_B T}} - A\,e^{-\frac{E_1}{K_B T}}$$

by S.G Sir

or, $$i = A\,e^{-\frac{(E_2 + |e|V_c - |e|V)}{K_B T}} - A\,e^{-\frac{E_1}{K_B T}}$$

or, $$i = A\,e^{-\frac{(E_1 - |e|V_c)}{K_B T}} - A\,e^{-\frac{E_1}{K_B T}}$$

where, $E_1 = E_2 + |e|V_c$ is raised barrier potential energy for hole in P-region.

or, $$i = A\,e^{-\frac{E_1}{K_B T}}.[e^{\frac{|e|V_C}{K_B T}} - 1]$$

or, $$i = i_0.[e^{\frac{|e|V_C}{K_B T}} - 1] \quad \ldots\ldots\ldots\ldots\ldots\ldots 4$$

where, $i_0 = A\,e^{-\frac{E_1}{K_B T}}$ is the current associated with flow of electrons (minority carriers) from P to N-region called reverse saturation current.

Hence, eqn 4 is the required expression for a current flowing across junction of diode at given temperature with V_C potential difference applied across it. This equation is also known as diode equation.

Voltage current (V-I) characteristics of P-N junction Diode:-

The voltage and current (V-I) characteristics of P-N junction diode shows the relation between bias potential and circuit current flowing across a junction diode. The characteistics are usually studied under two heads.

a. Forward bias characteristics:-

For forward bias voltage current characteristics, P-N junction diode is connected in forward bias potential with source battery along with milli ammeter (mA) in series and milli voltmeter (mV) across power supply as shown in fig,

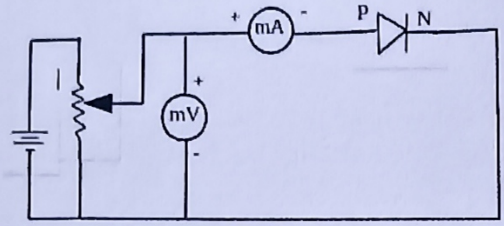


Fig: Circuit diagram of forward bias of junction diode

The forward bias voltage is increased gradually from zero value and resulting current across junction diode is recorded by milli ammeter (mA). The variation of forward current in diode with potential is as shown in fig;

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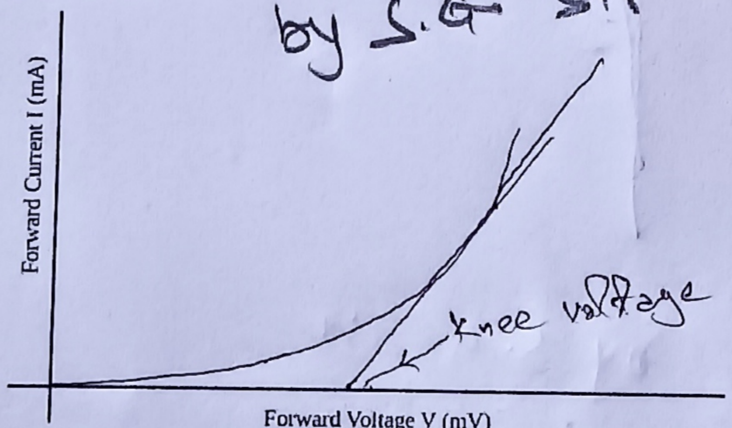

Fig: Forward bias voltage current characteristics of junction diode.

The plot shows that;

1. Initially, current in diode is zero when forward voltage is zero .
2. The forward current increases slowly up to certain voltage called Knee voltage. Keen voltage is the threshold or cut off voltage above which a current increases sharply across junction diode. The value of knee voltage is about 0.3 V for germanium and 0.7 V for silicon based diode.
3. After, knee voltage current in diode increases rapidly for small potential increase and curve is almost linear.

4. The collision of holes and electrons with +Ve crystal ions gives rise to ohmic dynamic resistance of diode. This limits amount of current flowing in diode. When excess current is allow to flow through diode, it will burn out due to more heating effect.

b. Reverse bias characteristics:-

For reverse bias voltage current characteristics, P-N junction diode is connected in reverse bias potential with source battery along with milli ammeter (mA) in series and milli voltmeter (mV) across power supply as shown in fig,

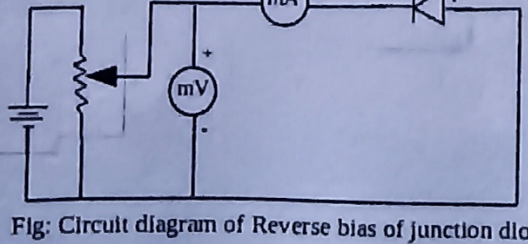

Fig: Circuit diagram of Reverse bias of junction diode.

The reverse bias voltage is increased gradually from zero value and resulting current across junction diode is recorded by milli ammeter (mA). The variation of reverse current in diode with potential is as shown in fig;

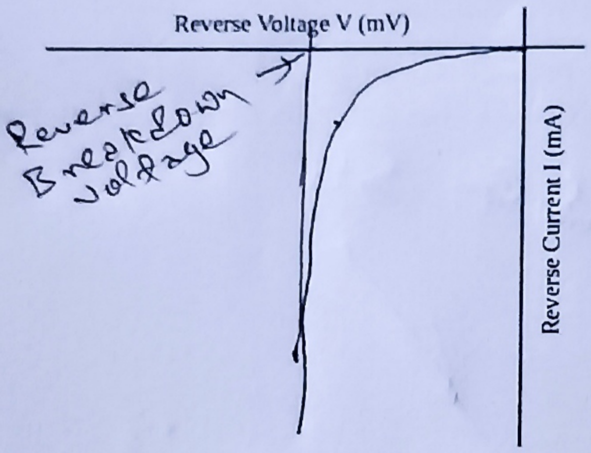

Fig: Reverse bias voltage current characteristics of junction diode.

by S. G Sir

The plot shows that;

1. When reverse voltage is increased from zero value, hence, reverse current (I_R) in diode increases and reaches to almost maximum value of current I_0 called reverse saturation current. This current in diode is due to minority charge carriers about junction of diode.
2. When on further increased on reverse voltage, at particular voltage, reverse current in diode suddenly and sharply increased. This voltage is called breakdown voltage.
3. On further increasing voltage beyond breakdown, diode likely to burn out due to high current and overheating.
4. The resistance of diode in reverse biasing is very large before breakdown voltage and drops to about zero beyond it.

Note: The large current in diode after breakdown is due to phenomenon named avalanche effect. When reverse potential is increased sufficiently, the minority charge carriers about junction gets accelerated and collide with atoms in their way and hence ejecting an electron from it. This process is multiplied in geometric progression resulting large free electron or current inside diode. Such effect in diode on its reverse bias condition is called avalanche breakdown effect.